Proto-Child as a Lawful Intermediate Support Mode in the Mesoscape Program

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Status

Working program note based on sandbox and probe results.

Claim

A persistent one-sided proto-child mode exists in the current mesoscape program as a lawful intermediate between no support and full support.

In the tested recurrent niche

$$[2,3] \to 5,$$

admitting this mode eliminates repeated same-child birth/retire cycling and redirects later local adjustment into weaken-mediated shell/interface re-expression.

1 Background

Earlier niche probes indicated that the recurrent niche

$$[2,3] \to 5$$

was not best understood as false novelty or bookkeeping noise. Instead, it behaved like a real local partial-relief demand.

The empirical pattern was:

1. births into the niche repeatedly showed strong parent relief and positive local differentiated-role gain,
2. full support occupancy in that niche later looked redundant enough that retirement improved structural distinctness and removed redundancy,
3. the partial-mode probe therefore favored a *sub-full burden-sharing interpretation* over shell redistribution or shared support as the primary missing mode.

This suggested that the missing object was not simply another full child, but some lawful intermediate support mode.

2 Sandbox construction

The proto-child sandbox introduced a semi-instantiated, one-sided child carrier. In the tested realization:

- child 5 was activated,
- the child was given reduced local amplitude,
- the child was attached only to parent 2,
- the second parent-child interface was not instantiated on the same move.

So the proto-child was neither:

- absence of support,
- nor a full two-parent child support patch.

It was an explicit intermediate support object.

3 Observed result

In the targeted run, the proto move at niche

$$[2,3] \to 5$$

was accepted at step 40 with strong support:

$$\begin{aligned} \Delta F &\approx 1.1019, \\ \text{proto\_readiness} &= 0.8, \\ \Delta O_{\text{diff}} &\approx 0.2323. \end{aligned}$$

At the run level, the sandbox produced:

Quantity	Value
Proto events	1
Birth events	6
Weaken events	22
Extinction events	1

This is qualitatively different from the earlier regime in which the same niche repeatedly expressed itself through same-child birth/retire cycling.

4 Stabilization result

The proto stabilization probe gives the decisive read:

`stable_proto_child_with_post_proto_weaken_flow`.

For the target niche

$$[2,3] \to 5,$$

the probe found:

1. the proto move was accepted at step 40,
2. child 5 remained present for 100% of post-proto snapshots,
3. post-proto same-child births were 0,
4. post-proto same-child retires were 0,
5. post-proto weakens were 22,
6. all recorded post-proto weakens were parent-incident and flagged as lawful shell re-expression,
7. in the final state, child 5 remained present and was attached only through edge $[2, 5]$.

So the old same-child birth/retire loop did not merely weaken. It disappeared.

5 Interpretation

The most natural interpretation is:

> The recurrent niche was not best modeled by repeated full birth followed by repeated retirement. It was better modeled by a stable one-sided proto-child carrier that lawfully resolves the local partial-relief demand, after which further local adjustment is handled through weaken-mediated shell/interface re-expression.

In that sense, proto-child behaves like a lawful mesoscopic intermediate between:

- no support,
- and full two-parent child support.

This result is consistent with the broader organizer and witness program, in which function-first persistence, lawful re-expression, and interface-centered relief matter more than raw graph motifs or raw support counts.

6 What this does show

This result supports the following statements:

1. A lawful intermediate support mode can exist in the current mesoscape setting.
2. That mode can stabilize a niche previously expressed as repeated same-child churn.
3. Once stabilized, the system can hand the remaining local adjustment over to weakening.
4. The resulting persistent object can remain asymmetric rather than consolidating immediately into a full two-parent support patch.

7 What this does not yet show

This result does *not* yet show:

1. broad healthy churn below capacity,
2. a general proto-child law for all niches,
3. a final particle-like interpretation,
4. that proto-child should already be promoted into the main law without further discipline.

So this is strong evidence for a new lawful intermediate, but not yet a final theory of intermediate support structure.

8 Programmatic consequence

Proto-child should no longer be treated as a disposable sandbox curiosity. It now deserves consideration as a disciplined lawful intermediate in the move program.

The next code step should therefore not be another ad hoc sandbox exception. Instead, the program should now think through how proto-child can be folded into the lawful move system in a controlled way.

That requires answering at least the following questions.

1. Admission condition

Under what witness conditions is proto-child allowed instead of full birth?

2. Identity condition

What makes a proto-child the same persistent local mode across later lawful moves?

3. Promotion / consolidation condition

Under what conditions does a proto-child lawfully become a full child?

4. Retirement / demotion condition

Under what conditions does a proto-child lawfully disappear or lawfully re-express into shell/interface weakening?

5. Bookkeeping condition

How should the organizer witness bundle represent proto-child support without collapsing into graph-only convenience?

9 Recommended next implementation target

The clean next implementation target is:

> Add proto-child as an explicitly named candidate move class in bookkeeping and driver logic, with witness-based admission and consolidation rules, rather than keeping it as a sandbox-only special case.

That would let the main program test whether proto-child is:

- a narrow niche fix,
- or a genuine lawful intermediate support mode of the mesoscape dynamics.

One-sentence summary

A persistent one-sided proto-child mode now appears to be a real lawful intermediate between no support and full support in the mesoscape program: in the tested niche

$$[2, 3] \to 5,$$

admitting it removed repeated same-child birth/retire cycling and redirected subsequent local adjustment into weaken-mediated shell/interface re-expression.